Software Requirements Specification (SRS)

Project Name: GREEN (Smart Community Farm System)

1. Introduction

The **Green Project** is a web-based application designed to manage community farming activities. It simplifies the process of renting agricultural plots, borrowing farming tools, and organizing volunteer work. The system serves as a bridge between land owners, farmers, and the community.

2. User Classes (Actors)

- **Farmer:** The main user who interacts with the system to rent land, borrow equipment, and participate in farm activities.
- **Admin:** The system manager who handles user approvals, inventory management, and monitors farm operations.

3. Functional Requirements

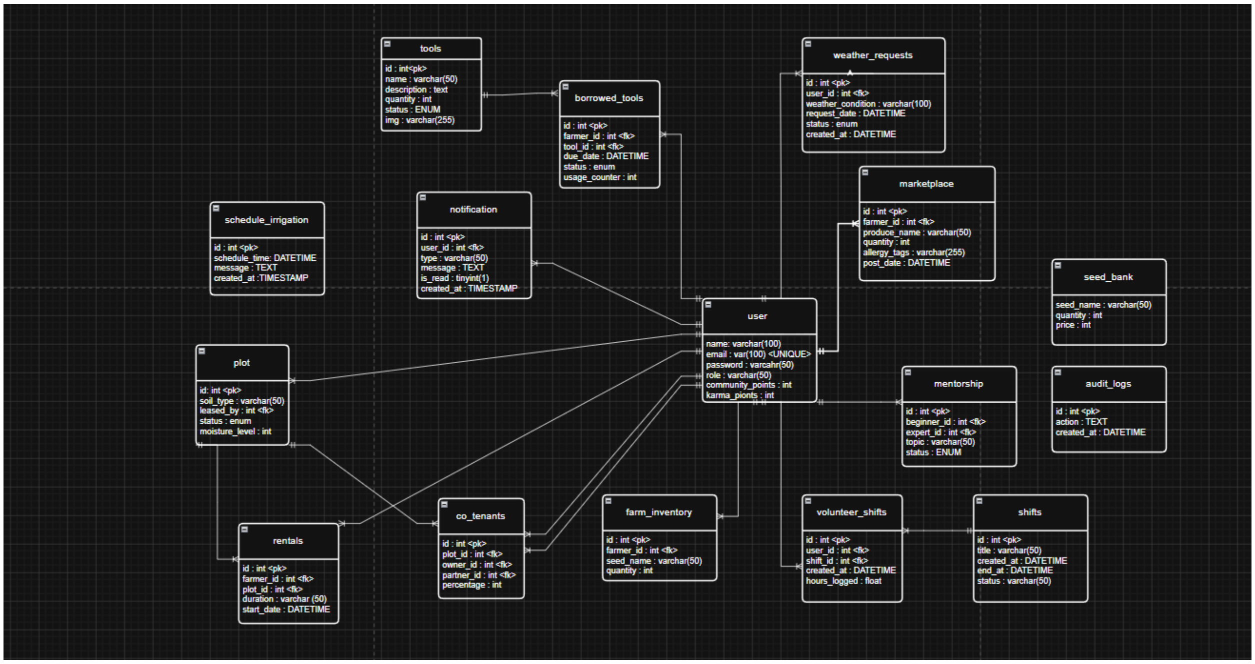
The system covers 28 Use Cases, organized into major modules:

- **Plot Management:** (Rent Plot), (View Plots), (Renew Rental).
- **Resource Management:** (Borrow Tool), (Return Tool), (Seed Bank Management).
- **Community Services:** (Mentorship), (Volunteer Shifts), (Pest Reporting).
- **Security:** (User Authentication/Login).

4. System Design (Diagrams)

4.1 Entity Relationship Diagram (ERD)

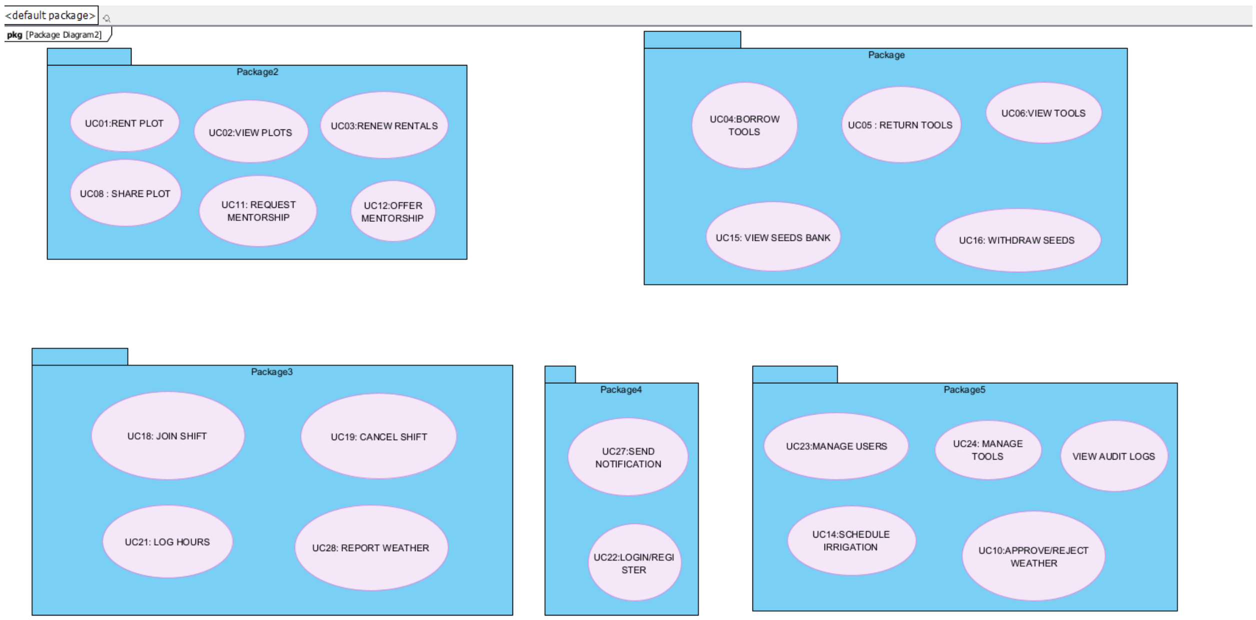
This diagram represents the database structure. It shows how the **Users** table connects to **Rentals**, **Tools**, and **Shifts**.



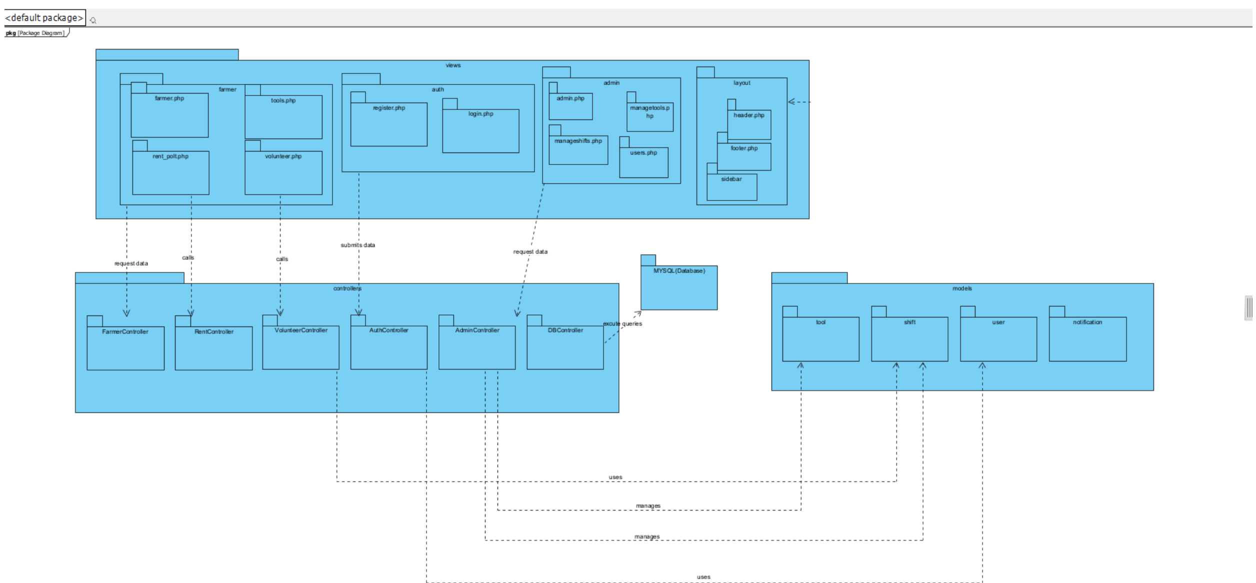
4.2 Class & Package Diagram

The system follows the **MVC (Model-View-Controller)** pattern.

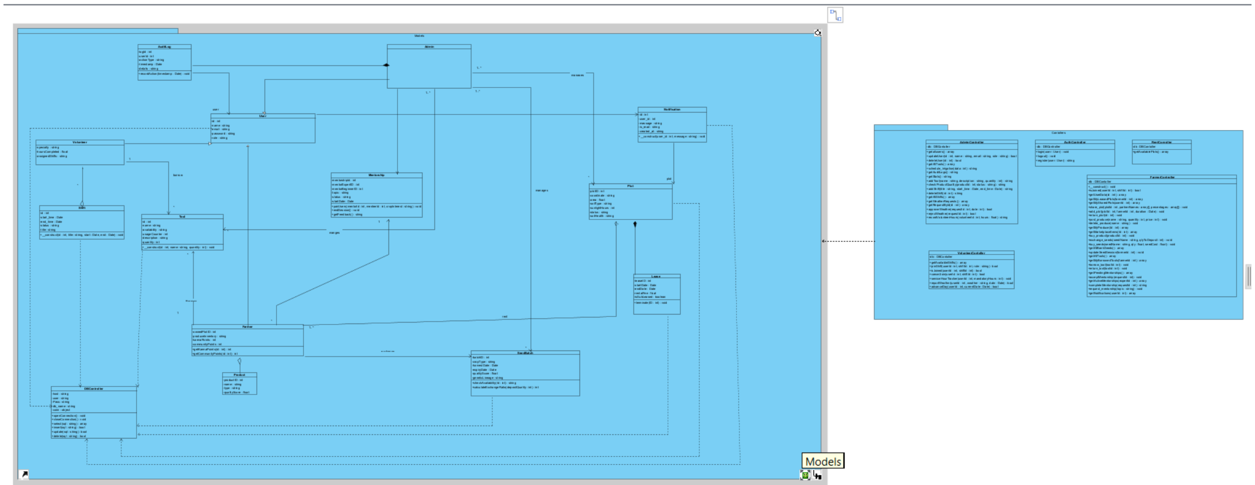
- **Packages:** Controllers (Logic), Models (Data), and Views (UI).
- **Classes:** Includes `FarmerController`, `AdminController`, and `DBController`.



<default package>
pkg [Package Diagram2]
Package2
UC01:RENT PLOT
UC02:VIEW PLOTS
UC03:RENEW RENTALS
UC08 : SHARE PLOT
UC11: REQUEST MENTORSHIP
UC12:OFFER MENTORSHIP
Package
UC04:BORROW TOOLS
UC05 : RETURN TOOLS
UC06:VIEW TOOLS
UC15: VIEW SEEDS BANK
UC16: WITHDRAW SEEDS
Package3
UC18: JOIN SHIFT
UC19: CANCEL SHIFT
UC21: LOG HOURS
UC28: REPORT WEATHER
Package4
UC27:SEND NOTIFICATION
UC22:LOGIN/REGISTER
Package5
UC23:MANAGE USERS
UC24: MANAGE TOOLS
VIEW AUDIT LOGS
UC14:SCHEDULE IRRIGATION
UC10:APPROVE/REJECT WEATHER



<default package>
pkg [Package Diagram]
views
farmer
farmer.php
tools.php
rent_polt.php
volunteer.php
auth
register.php
login.php
admin
admin.php
managetools.php
manageshifts.php
users.php
layout
header.php
footer.php
sidebar
request data
calls
calls
submits data
request data
MYSQL(Database)
controllers
FarmerController
RentController
VolunteerController
AuthController
AdminController
DBController
excute queries
models
tool
shift
user
notification
uses
manages
manages
uses



5. Non-Functional Requirements

- **Security:** Unauthorized users cannot access the farmer or admin dashboards.
- **Performance:** The system should update the status of tools and plots in real-time.
- **Availability:** The system must be accessible for farmers to check their schedules at any time.